

Practice Quiz: Learning (Philosophical Foundations)

Name: __ Date: __

Part A: Multiple Choice

1. Kuhn's paradigm shifts map onto Active Inference as: A) Random model changes B) Bayesian model selection — a new model is chosen when it has higher model evidence than the old one C) Forgetting all parameters D) Increasing complexity indefinitely
2. Occam's Razor is formalized in Active Inference through: A) Ignoring complex models B) The free energy bound $F = \text{Complexity} - \text{Accuracy}$, which penalizes unnecessary model complexity C) Always choosing the simplest model regardless of accuracy D) The law of parsimony in Latin
3. Bayesian falsification differs from Popperian falsification because: A) It's less rigorous B) It is gradual — model evidence declines relative to competitors rather than a single observation refuting a theory C) It ignores evidence D) It only works with quantum mechanics
4. In Active Inference terms, pseudoscience is characterized by: A) Mathematical models B) Models that maximize complexity (explain everything post-hoc) without generating testable predictions C) Using computers D) Having many followers
5. The problem of induction is addressed by Active Inference through: A) Logical proof B) Pragmatic justification — inductive priors "work" because agents with bad priors don't survive C) Ignoring the problem D) Assuming the future equals the past
6. Bayesian Model Reduction (BMR) implements: A) Model expansion B) Active simplification — pruning parameters that don't improve model evidence C) Random deletion D) Adding hidden states
7. The FEP is closest to which philosophy of science position? A) Naïve realism B) Constructive empiricism — science aims at predictive adequacy rather than truth about unobservables C) Postmodernism D) Logical positivism

Part B: Essay Questions

1. Analyze a historical scientific revolution (your choice) through Active Inference. Map the phases of normal science, crisis, and revolution onto parameter learning, model evidence decline, and Bayesian model selection. Was the revolution "rational" in Bayesian terms? (300 words)
2. How does Active Inference account for persistent irrational beliefs (e.g., conspiracy theories, ideological rigidity)? Use the concepts of precision, prior strength, and confirmation bias to explain why some agents resist evidence. (300 words)
3. "Occam's Razor is not just a heuristic — it's a theorem." Evaluate this claim using the free energy decomposition $F = \text{Complexity} - \text{Accuracy}$. Under what conditions does the simplest model win? When might Occam's Razor fail? (300 words)